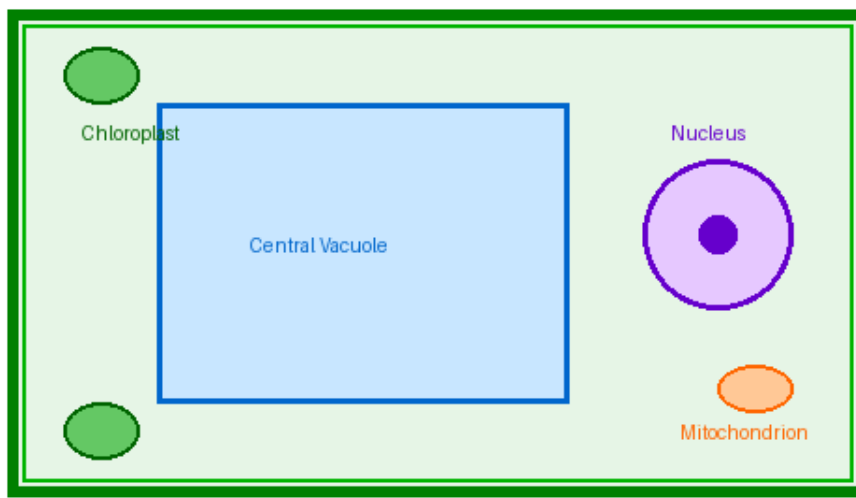


Study Notes: Biology Essentials

Unit 1: Cell Biology and Organelles

Cells are the structural and functional units of all living organisms. Plant cells contain unique organelles that differentiate them from animal cells. Most notably, plant cells have a rigid outer cell wall composed of cellulose which provides structural support, and chloroplasts containing chlorophyll for harvesting sunlight. They also feature a large central vacuole which stores water and maintains turgor pressure within the cell.

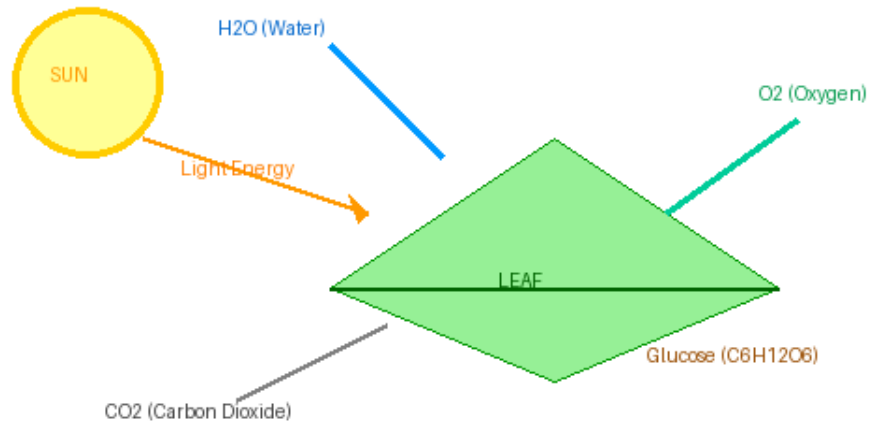
Figure 1: Structure of a Plant Cell



Unit 2: Photosynthesis and Energy Conversion

Photosynthesis is the metabolic process by which photoautotrophs convert light energy into chemical energy. This biological process occurs inside the chloroplasts of plant leaves. During this reaction, carbon dioxide and water are consumed in the presence of light to synthesize glucose and release oxygen. The balanced chemical equation is represented as: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

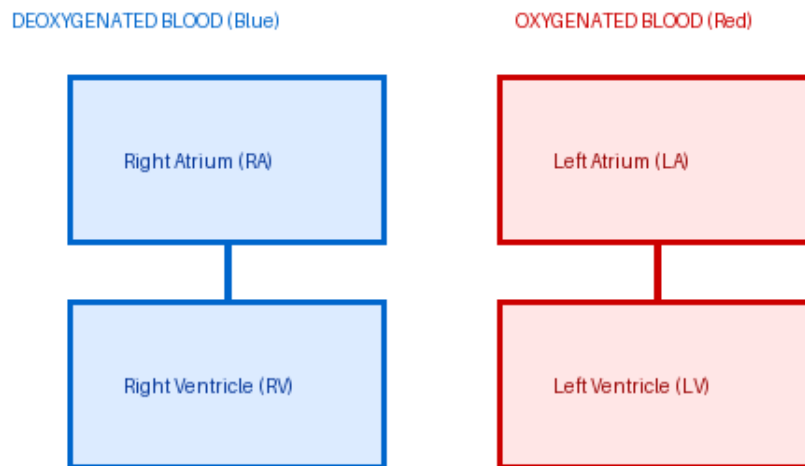
Figure 2: Photosynthesis Reaction Pathways



Unit 3: The Human Circulatory System

The human circulatory system is responsible for transporting oxygen, nutrients, and waste throughout the body. The heart serves as the muscular pump of this system and is divided into four distinct chambers: the Right Atrium, Right Ventricle, Left Atrium, and Left Ventricle. Deoxygenated blood returns from the body to the right side of the heart, which pumps it to the lungs. Oxygenated blood returns from the lungs to the left side of the heart, which distributes it to the body tissues.

Figure 3: Four Chambers of the Human Heart



Unit 4: Mendelian Genetics & Genotypes

Gregor Mendel established the laws of inheritance by studying pea plants. A monohybrid cross examines the inheritance patterns of a single trait. A Punnett Square is a graphical representation used to predict the genotypes of offspring from a cross. By aligning maternal and paternal alleles (e.g., T for dominant tall allele and t for recessive dwarf allele), we can calculate the probability of homozygous and heterozygous offspring.

Figure 4: Monohybrid Cross Punnett Square

